Discrete and Continuous Random Variable

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Random Variables

Concept of a Random Variable:

A variable whose (numerical) value is determined by the outcome of a random experiment is called a random variable.

Example 1:

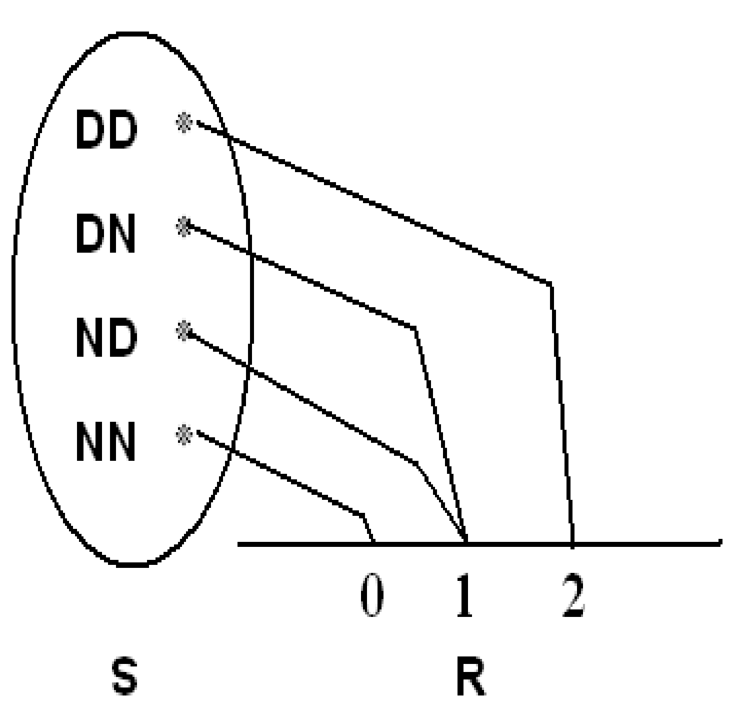
Experiment: testing two components.

(D=defective, N=non-defective)

Sample space: $S = \{DD, DN, ND, NN\}$

Let X = number of defective components when two components are tested.

Assigned numerical values to the outcomes are:



Notice that, the set of all possible values of the random variable X is {0, 1, 2}.

Definition:

A random variable X is a function that associates each element in the sample space with a real number (i.e., X : $S \to$ **R**.)

Notation: " X " denotes the random variable ." x " denotes a value of the random variable X.

Types of Random Variables:

- A random variable X is called a **discrete** random variable if its set of possible values is countable, i.e.,

$$x \in \{x_1, x_2, \dots, x_n\} \text{ or } x \in \{x_1, x_2, \dots\}$$

- A random variable X is called a **continuous** random variable if it can take values on a continuous scale, i.e.,

$$x \in \{x: a < x < b;\ a, b \in R\}$$

Discrete Probability Distributions

Definition: If X is a discrete random variable the function given by f(x) = P(X = x) for each x within the range of X is called the probability distribution of X.

Theorem 1:

A function can serve as the probability distribution of the discrete random variable X if and only if its value , f(x), satisfy the conditions.

$$1- \ f(x) \geq 0 \qquad\qquad 2- \ \sum_{x} f(x) = 1$$

Example 2*:*

Experiment: tossing a non-balance coin 2 times independently such that P(H) = 1/2 P(T) find the probability distribution of number of heads:

Solution

Let X= number of heads

$\because P(H)+P(T)=1$ and $P(H)=\frac{1}{2}P(T)$

$\therefore P(H)=\frac{1}{3}$ $P(T)=\frac{2}{3}$

Sample space: S ={HH, HT, TH, TT}

Sample point (Outcome)	Probability	Value of (x)
HH	$P(HH)=P(H)\,P(H)=1/3\times1/3 = 1/9$	2
HT	$P(HT)=P(H)\,P(T)=1/3\times2/3 = 2/9$	1
TH	$P(TH)=P(T)\,P(H)=2/3\times1/3 = 2/9$	1
TT	$P(TT)=P(T)\,P(T)=2/3\times2/3 = 4/9$	0

Then the probability distribution of X is:

X	0	1	2	Total
$f(x) = P(X=x)$	4/9	4/9	1/9	1

Example 3:

For the previous example, find:

$P(X < 1)$, $P(X \leq 1)$, $P(X \geq 0.5)$, $P(X > 8)$, $P(X < 10)$.

X	0	1	2	Total
$f(x) = P(X=x)$	4/9	4/9	1/9	1

Solution

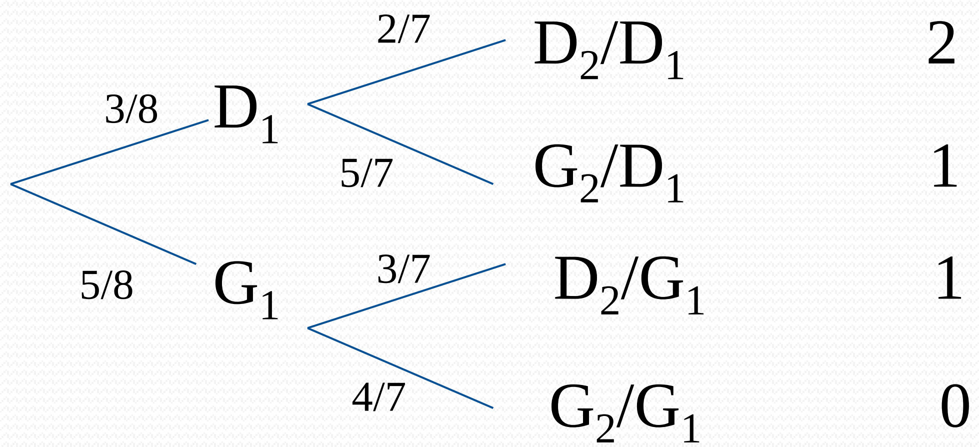
- $P(X < 1) = P(X = 0) = 4/9$
- $P(X \leq 1) = P(X = 0) + P(X = 1) = 4/9 + 4/9 = 8/9$
- $P(X \geq 0.5) = P(X = 1) + P(X = 2) = 4/9 + 1/9 = 5/9$
- $P(X > 8) = P(\phi) = 0$
- $P(X < 10) = P(X = 0) + P(X = 1) + P(X = 2) = P(S) = 1$

Example 4:

A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective and 5 are non-defective. If a school makes a random purchase of 2 of these computers, find the probability distribution of the number of defectives.

Solution

We need to find the probability distribution of the random variable: X = the number of defective computers purchased.



The possible values of X are: x = 0, 1, 2.

$$f(0) = P(X = 0) = \frac{5}{8} \times \frac{4}{7} = \frac{10}{28}$$

$$f(1) = P(X = 1) = \frac{5}{8} \times \frac{3}{7} + \frac{3}{8} \times \frac{5}{7} = \frac{15}{28}$$

$$f(2) = P(X = 2) = \frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$$

In general, for x = 0, 1, 2, we have:
The probability distribution of X is:

X	**0**	**1**	**2**	**Total**
f(x) = P(X = x)	10/28	15/28	3/28	1

Definition:

The Cumulative Distribution Function (CDF), F(x), of a discrete random variable X with the probability function f(x) is given by:

$$F(x) = P(X \le x) = \sum_{t \le x} f(t) = \sum_{t \le x} P(X = t) \quad \text{for} \quad -\infty < x < \infty$$

Theorem 2:

The value F(x) of the distribution function of a discrete r.v. X satisfy the conditions

- $F(-\infty) = 0$ and $F(\infty) = 1$
- If $a \le b$, the $F(a) \le F(b)$ for any real numbers a and b.

Example 5:

Find the CDF of the random variable X with the probability function:

X	0	1	2	Total
$f(x) = P(X = x)$	10/28	15/28	3/28	1

Solution:

$F(x) = P(X \leq x)$ for $-\infty < x < \infty$

For $x < 0$: $F(x) = 0$

For $0 \leq x < 1$: $F(x) = P(X = 0) = 10/28$

For $1 \leq x < 2$: $F(x) = P(X = 0) + P(X = 1)$

$= 10/28 + 15/28 = 25/28$

For $x \geq 2$: $F(x) = P(X = 0) + P(X = 1) + P(X = 2)$

$= 10/28 + 15/28 + 3/28 = 28/28 = 1$

The CDF of the random variable X is:

$$F(x) = \begin{cases} 0 & ;\ x < 0 \\ \dfrac{10}{28} & ;0 \leq x < 1 \\ \dfrac{25}{28} & ;1 \leq x < 2 \\ 1 & ;x \geq 2 \end{cases}$$

Note:

For a discrete random variable X, we have:

- $P(x \le a) = F(a)$
- $P(a < x \le b) = F(b) - F(a)$

Example 6

$P(X \le -0.5) = F(-0.5) = 0$

$P(X \le 1.5) = F(1.5) = F(1) = 25/28$

$P(X \le 3.8) = F(3.8) = F(2) = 1$

$P(0.5 \le X \le 1.5) = F(1.5) - F(0.5) = 25/28 - 10/28 = 15/28$

Example 7:

If the distribution function o X is given by

$$F(x)=\begin{cases} 0 & ; x<0 \\ 1/4 & ;0\le x<1 \\ 1/2 & ;1\le x<2 \\ 5/8 & ;2\le x<3 \\ 3/4 & ;3\le x<4 \\ 7/8 & ;4\le x<5 \\ 1 & ;x\ge 5 \end{cases}$$

Find the probability distribution of the r.v.

Solution

Note:

For a discrete random variable X, we have:

- $P(x \le a) = F(a)$
- $P(a \le x < b) = F(b) - F(a)$

f(0) =F(0) = 1/4, *f(1) = F(1) - F(0) = 1/4.*

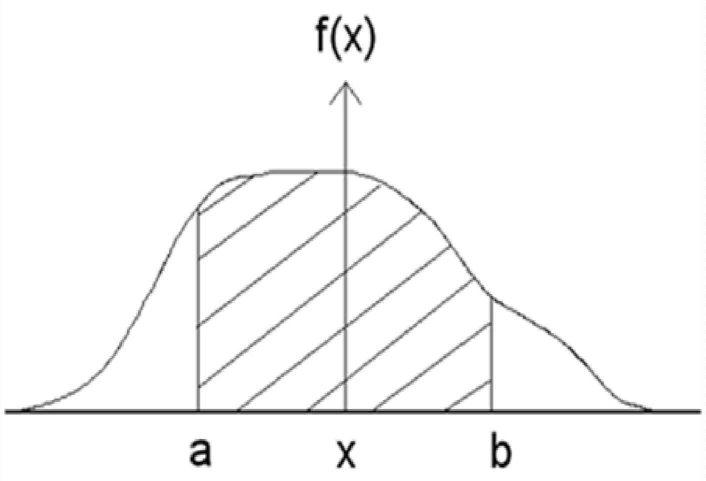
f(2) = F(2) - F(1) = 1/8 , *f(3) = F(3) - F(2) = 1/8.*

f(4) = F(4) - F(3) = 1/8, *f(5) = F(5) - F(4) = 1/8.*

$$f(x) = \begin{cases} 1/4 & ;\ x = 0, 1 \\ 1/8 & ;\ x = 2, 3, 4, 5 \\ 0 & ;\ o.w. \end{cases}$$

Continuous Probability Distributions

For any continuous random variable, X, there exists a non-negative function f(x), called the probability density function (p.d.f) through which we can find probabilities of events expressed in term of X.



$f: R \to [0, \infty)$

$P(a < X < b) =$

= area under the curve of f(x) and over the interval (a,b)

$P(X \in A) =$

= area under the curve of f(x) and over the region A

Theorem :

The function f(x) is a probability density function (p.d.f) for a continuous random variable X, defined on the set of real numbers, if and only if:

$$1- \qquad f(x) \geq 0 \qquad \forall\, x \in R$$

$$2- \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1$$

Note:

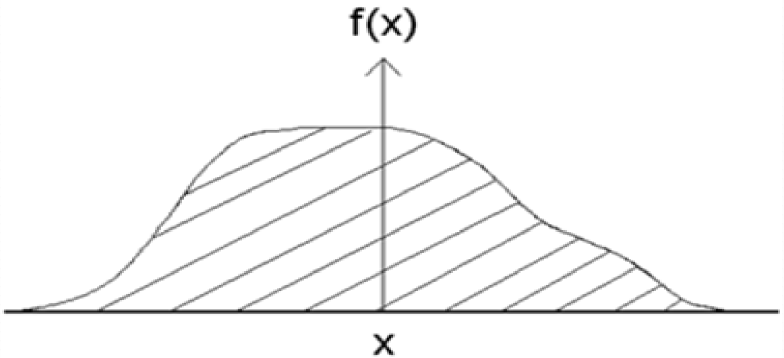
For a continuous random variable X, we have:

1- $f(x) \neq P(X = x)$ (in general)

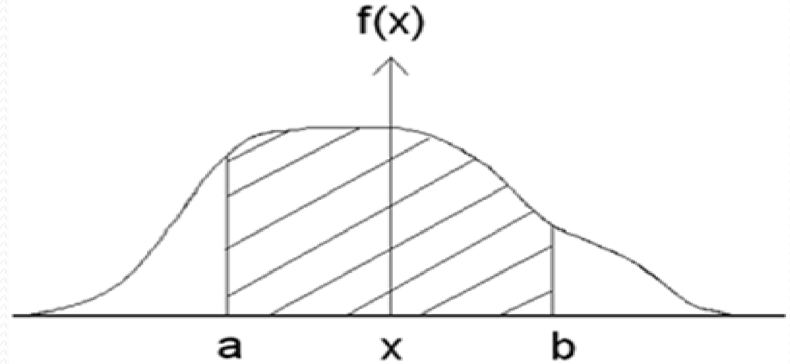
2- $P(X = a) = 0$ for any $a \in R$

3- $P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b)$

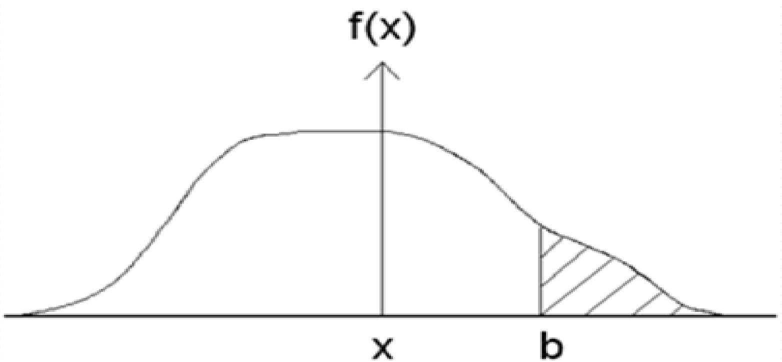
$$= P(a < X < b) = \int_a^b f(x)\,dx$$



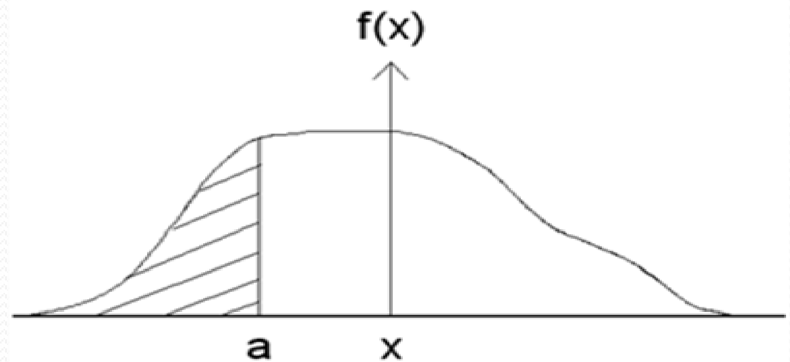
$$\int_{-\infty}^{\infty} f(x)dx = 1$$



$$P(a < X < b) = \int_{a}^{b} f(x)dx$$



$$P(X > b) = \int_{b}^{\infty} f(x)dx$$



$$P(X < a) = \int_{-\infty}^{a} f(x)dx$$

Example 1*:*

Suppose that the error in the reaction temperature, in C^o, for a controlled laboratory experiment is a continuous random variable X having the following probability density function:

$$f(x)=\begin{cases}\frac{1}{3}x^2 \;;\; -1<x<2\\ 0\;;\; elsewhere\end{cases}$$

Verify that

(a) $f(x) \geq 0$

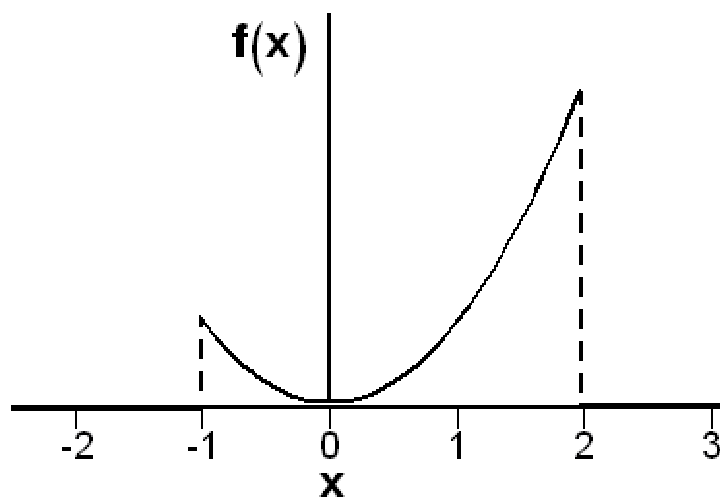
(b) $\int_{-\infty}^{\infty} f(x)dx = 1$

(c) And find $P(0 < X \leq 1)$

Solution:

X = the error in the reaction temperature in Co. X is continuous r. v.

$$f(x)=\begin{cases}\frac{1}{3}x^2 \ ; -1<x<2\\ 0 \ ; \ elsewhere\end{cases}$$

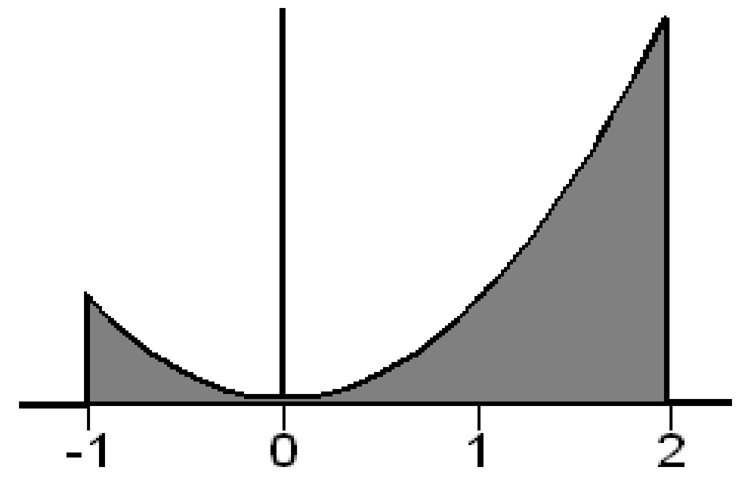


(a) f(x) ≥ 0 because f(x) is a quadratic function.

(b) $\int_{-\infty}^{\infty} f(x)dx = 1$

$$\int_{-\infty}^{\infty} \mathrm{f(x)\,dx} = \int_{-\infty}^{-1} 0\,\mathrm{dx} + \int_{-1}^{2} \frac{1}{3}\mathrm{x}^2\,\mathrm{dx} + \int_{2}^{\infty} 0\,\mathrm{dx}$$

$$= \int_{-1}^{2} \frac{1}{3} x^2 \, dx = \left[\frac{1}{9} x^3 \;\middle|\; \begin{matrix} x = 2 \\ x = -1 \end{matrix} \right]$$

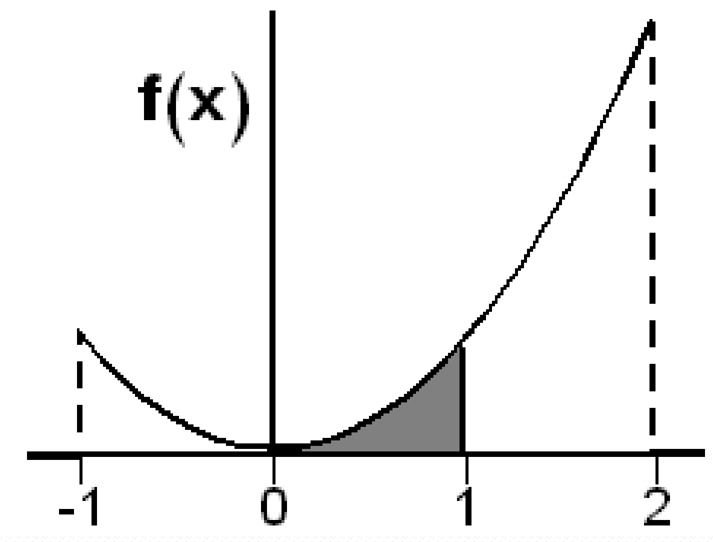


$$= \frac{1}{9}(8 - (-1)) = 1$$

$$\text{(c) } P(0 < X \leq 1) = \int_{0}^{1} f(x) \, dx = \int_{0}^{1} \frac{1}{3} x^2 \, dx$$

$$= \left[\frac{1}{9} x^3 \;\middle|\; \begin{matrix} x = 1 \\ x = 0 \end{matrix} \right]$$

$$= \frac{1}{9}(1 - (0)) = \frac{1}{9}$$



Example 2:

Suppose that X has a probability density function given by

$$f(x) = \begin{cases} k\,x\,(1-x) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the value of k.

(b) Find P(0.5 < X < 0.8).

Solution

$$1- \quad \because \int_0^1 f(x) = 1$$

$$\int_0^1 kx(1-x) = k\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{x=0}^{x=1} = \frac{1}{6}k = 1$$

$$\therefore k = 1$$

$$2- \quad P(0.5 < X < 0.8) = \int_{0.5}^{0.8} 6x(1-x)$$

$$= 6\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{x=0.5}^{x=0.8} = 0.396$$

Definition:

The cumulative distribution function (CDF), F(x), of a continuous random variable X with probability density function f(x) is given by:

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)dx \qquad \text{for} \qquad -\infty < x < \infty$$

Result:

For any real number a and b with $a \le b$

1- $P(a < X \le b) = P(X \le b) - P(X \le a) = F(b) - F(a)$

2- $f(x) = \dfrac{dF(x)}{dx}$

Example 3:

in Example 1,

1- Find the CDF

2- Using the CDF, find $P(0 < X \le 1)$.

Solution:

$$f(x) = \begin{cases} \frac{1}{3}x^2 \; ; -1 < x < 2 \\ 0 \; ; \; elsewhere \end{cases}$$

For x < –1:

$$F(x)=\int_{-\infty}^{x} f(t)dt = \int_{-\infty}^{x} 0\,dt = 0$$

For –1 ≤ x < 2:

$$F(x)=\int_{-\infty}^{x} f(t)dt = \int_{-\infty}^{-1} 0\,dt + \int_{-1}^{x} \frac{1}{3}t^2\,dt = \int_{-1}^{x} \frac{1}{3}t^2\,dt$$

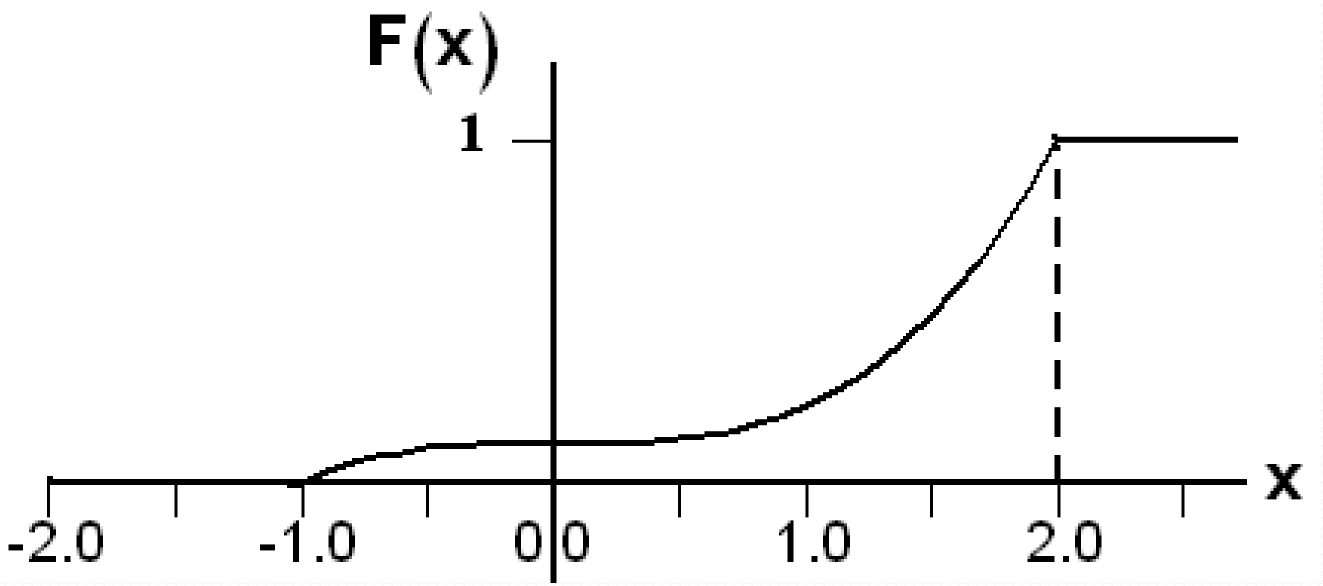
$$=\left[\frac{1}{9}t^3\right]_{t=-1}^{t=x} = \frac{1}{9}\left(x^3-(-1)\right) = \frac{1}{9}\left(x^3+1\right)$$

For x ≥ 2:

$$F(x)=\int_{-\infty}^{x} f(t)dt = \int_{-\infty}^{-1} 0\,dt + \int_{-1}^{2} \frac{1}{3}t^2\,dt + \int_{2}^{x} 0\,dt = \int_{-1}^{2} \frac{1}{3}t^2\,dt = 1$$

Therefore, the CDF is:

$$F(x) = P(X \le x) = \begin{cases} 0 & ; x < -1 \\ \dfrac{1}{9}(x^3 + 1) & ; -1 \le x < 2 \\ 1 & ; x \ge 2 \end{cases}$$



2. Using the CDF,

$$P(0 < x < 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}$$

Example 4:

If X has the probability density

$$f\left(x\right)=\begin{cases} K\,e^{-3x} & ;\ x>0 \\ 0 & ;\ o.w. \end{cases}$$

1- Find K ,

2- $P(0.5 < x \le 1)$

3- F(x).

4- by using F(x) find $P(0 < x < 0.3)$

Solution

1- From the second condition of the theorem 3

$$\int_{-\infty}^{\infty} f(x)dx = \int_{0}^{\infty} Ke^{-3x}\,dx = \left[K.\frac{e^{-3x}}{-3}\right]_0^{\infty} = \frac{K}{3} = 1$$

$$Then \quad K = 3$$

$$2 - P(0.5 \le x \le 1) = \int_{0.5}^{1} 3e^{-3x}\,dx = \left[-e^{-3x}\right]_{0.5}^{1}$$

$$= -e^{-3} + e^{-1.5} = 0.173$$

3- For x $\leq$ 0 we can write F(x) = 0

For x > 0

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{0}^{x} 3e^{-3x}\,dx$$

$$= \left[-e^{-3x}\right]_0^x = 1 - e^{-3x}$$

Then

$$F(x) = \begin{cases} 1 - e^{-3x} & ;\ x > 0 \\ 0 & ;\ x \leq 0 \end{cases}$$

4- by using F(x) find P(0 < x < 0.3)

$$P(0 < x < 0.3) = F\left(0.3\right) - F\left(0\right) = 1 - e^{-0.9} - 0 = 0.5934$$

Mathematical Expectation:

Mean of a Random Variable:

Definition 1:

Let X be a random variable with a probability distribution $f(x)$. The mean (or expected value) of X is denoted by μ_X (or $E(X)$) and is defined by:

$$E(X) = \mu_X = \begin{cases} \sum_{all\ x} x\, f(x)\ ; & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x\, f(x)\, dx\ ; & \text{if } X \text{ is continuous} \end{cases}$$

Example 1:

A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective and 5 are non-defective. If a school makes a random purchase of 2 of these computers, find the expected number of defective computers purchased

Solution:

Let X is the number of defective computers purchased.

In Example 4, (Lecture 5) we found that the probability distribution of X is:

X	0	1	2	Total
f(x) =P(X=x)	10/28	15/28	3/28	1

The expected value of the number of defective computers purchased is the mean (or the expected value) of X, which is:

x	***f(x) = P(X=x)***	***x.f(x)***
0	10/28	0
1	15/28	15/28
2	3/28	6/28
		$\sum xf(x) = 21/28$

$$E(x) = \sum x\, f(x) = 21/28 = 0.75$$

Example 2:

Let X be a continuous random variable that represents the life (in hours) of a certain electronic device. The *p.d.f* of X is given by:

$$f(x) = \begin{cases} \dfrac{20000}{x^3} & ; x > 100 \\ 0 & ; \quad \text{o.}w \end{cases}$$

Find the expected life of this type of devices.

Solution:

$$E(x) = \mu_x = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

$$= \int_{100}^{\infty} x \frac{20000}{x^3} dx = 20000 \int_{100}^{\infty} \frac{1}{x^2} dx$$

$$= 20000 \left[\frac{-1}{x} \right]_{x=100}^{x=\infty} = 20000 \left[0 - \frac{-1}{100} \right] = 200$$

Therefore, we expect that this type of electronic devices to last, on average, 200 hours.

Theorem 1:

Let X be a random variable with a probability distribution *f(x)*, and let *g(X)* be a function of the random variable *X*. The mean (or expected value) of the random variable *g(X)* is denoted by $\mu_{g(X)}$ (or *E[g(X)]*) and is defined by:

$$E[g(X)] = \mu_{g(X)} = \begin{cases} \sum_{all\ x} g(x) f(x)\ ; & if\ \ X\ \ is\ discrete \\ \int_{-\infty}^{\infty} g(x) f(x) dx\ ; & if\ \ X\ \ is\ continuous \end{cases}$$

Example 3:

Let X be a discrete random variable with the following probability distribution

X	0	1	2	Total
f(x) =P(X=x)	10/28	15/28	3/28	1

Find $E[g(x)]$ where, $g(x)=(x-1)^2$

Solution:

x	$g(x)=(x-1)^2$	$f(x)=P(X=x)$	$g(x)f(x)$
0	1	10/28	10/28
1	0	15/28	0
2	1	3/28	3/28
			$\sum g(x)f(x)=13/28$

$$E[g(x)]=\mu_{g(x)}=\sum_{x=0}^{x=2}(x-1)^2 f(x)=\frac{13}{28}$$

Example 4:

In Example 2,

$$f(x)=\begin{cases}\dfrac{20{,}000}{x^{3}} & ;\ x>100\\ 0\ ;\ elsewhere\end{cases}$$

find $\mathrm{E}\left(\dfrac{1}{\mathrm{X}}\right)$

Solution

$$\mathrm{E[g(X)]}=\mu_{\mathrm{g(X)}}=\int_{-\infty}^{\infty} g(x)\, f(x)\, dx=\int_{-\infty}^{\infty}\frac{1}{x} f(x)\, dx$$

$$\mathrm{E}\left(\frac{1}{\mathrm{X}}\right)=\int_{100}^{\infty}\frac{1}{x}\frac{20000}{x^{3}}dx=20000\int_{100}^{\infty}\frac{1}{x^{4}}dx$$

$$=\frac{20000}{-3}\left[\frac{1}{x^{3}}\Bigg|_{x=100}^{x=\infty}\right]=\frac{-20000}{3}\left[0-\frac{1}{1000000}\right]=0.0067$$

Variance (of a Random Variable):

The most important measure of variability of a random variable X is called the variance of X and is denoted by σ_X^2 or Var(X)

Definition 2:

Let X be a random variable with a probability distribution $f(x)$ and mean μ. The variance of X is defined by:

$$\mathrm{Var}(X) = \sigma_X^2 = E[(X-\mu)^2] = \begin{cases} \sum_{all\ x} (x-\mu)^2\ f(x)\ ; & if\ \ X\ \ is\ discrete \\ \int_{-\infty}^{\infty} (x-\mu)^2\ f(x)\,dx\ ; & if\ \ X\ \ is\ continuous \end{cases}$$

Definition 3:

The positive square root of the variance of *X*, is called the standard deviation of *X*.

Note:

$$Var(X) = E[g(X)] \quad \text{where} \quad g(X) = \left(X - \mu\right)^2$$

Theorem 2:

The variance of the random variable *X* is given by:

$$Var(X) = \sigma_X^2 = E\left(X^2\right) - \mu^2$$

Where

$$\mathrm{E}(\mathrm{X}^2) = \begin{cases} \sum_{all\ x} x^2 f(x)\,; & if\ X\ is\ discrete \\ \int_{-\infty}^{\infty} x^2 f(x)\,dx\,; & if\ X\ is\ continuous \end{cases}$$

Example 5:

Let X be a discrete random variable with the following probability distribution

X	0	1	2	3
f(x) =P(X=x)	0.51	0.38	0.10	0.01

Find $Var(X)$

Solution

x	f(x) =P(X=x)	$x\ f(x)$	$x^2 f(x)$
0	0.51	0	0
1	0.38	0.38	0.38
2	0.10	0.20	0.40
3	0.01	0.03	0.09
		$\sum xf(x) = 0.61$	$\sum x^2 f(x) = 0.87$

$$Var(X) = \sigma_X^2 = E(X^2) - \mu^2$$

$$= 0.87 - (0.61)^2 = 0.4979$$

Example 6:

Let X be a continuous random variable with the following p.d.f:

$$f(x)=\begin{cases}2(x-1)\ ;\ 1<x<2\\ 0\ ;\ elsewhere\end{cases}$$

Find the mean and the variance of X.

Solution:

$$\mu = E(X) = \int_{-\infty}^{\infty} x\, f(x)\,dx = \int_{1}^{2} x\left[2(x-1)\right]dx = \int_{1}^{2}\left(2x^2-2x\right)dx$$

$$= \left[2\frac{x^3}{3} - 2\frac{x^2}{2}\right]_1^2 = \left[\left(2\frac{8}{3}-4\right)-\left(2\frac{1}{3}-1\right)\right] = \frac{5}{3}$$

$$E\left(X^2\right) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_{1}^{2} x^2\left[2(x-1)\right]dx = \int_{1}^{2}\left(2x^3-2x^2\right)dx$$

$$= \left[2\frac{x^4}{4} - 2\frac{x^3}{3}\right]_1^2 = \left[\left(2\frac{16}{4}-2\frac{8}{3}\right)-\left(2\frac{1}{4}-2\frac{1}{3}\right)\right] = \frac{17}{6}$$

$$Var(X) = \sigma_X^2 = E\left(X^2\right) - \mu^2$$

$$= \frac{17}{6} - \left(\frac{5}{3}\right)^2 = \frac{1}{18}$$

Means and Variances of Linear Combinations of Random Variables:

If $X_1, X_2, \ldots, X_n$ are n random variables and $a_1, a_2, \ldots, a_n$ are constants, then the random variable :

$$Y = \sum_{i=1}^{n} a_i X_i = a_1 X_1 + a_2 X_2 + \cdots + a_n X_n$$

is called a linear combination of the random variables X_1, X_2, …, X_n.

Theorem 3:

If X is a random variable with mean $\mu = E(X)$, and if a and b are constants, then:

$$E(aX \pm b) = a\,E(X) \pm b$$

$$\mu_{\mathrm{aX\pm b}} = \mathrm{a}\mu_{\mathrm{X}} \pm \mathrm{b}$$

Corollary 1: $E(b) = b$ ($a = 0$ in Theorem 3)

Corollary 2: $E(aX) = a\,E(X)$ ($b = 0$ in Theorem 3)

Example 7:

Let X be a random variable with the following probability density function:

$$f(x)=\begin{cases}\frac{1}{3}x^2 \; ; -1<x<2 \\ 0 \; ; \; elsewhere\end{cases}$$

Find *E(4X+3)*.

Solution

$$\mu = E(X) = \int_{-\infty}^{\infty} x\, f(x)\,dx = \int_{-1}^{2} x \left[\frac{1}{3}x^2\right] dx = \int_{-1}^{2} \left(\frac{1}{3}x^3\right) dx$$

$$= \left[\frac{1}{12}x^4\right]_{-1}^{2} = \left(\frac{16}{12} - \frac{1}{12}\right) = \frac{15}{12} = \frac{5}{4}$$

$$E(4X+3) = 4\,E(X)+3 = 4(5/4) + 3 = 8$$

Another solution:

$$E\left[g(X)\right] = \int_{-\infty}^{\infty} g(X)\, f(x)\,dx = \int_{-1}^{2} (4x+3) \left[\frac{1}{3}x^2\right] dx$$

$$= \int_{-1}^{2} \left(\frac{4}{3}x^3 + x^2\right) dx = \left[\frac{x^4}{3} + \frac{x^3}{3}\right]_{-1}^{2}$$

$$= \left(\left(\frac{16}{3} + \frac{8}{3}\right) - \left(\frac{1}{3} + \frac{-1}{3}\right)\right) = 8 - 0 = 8$$

Theorem 4

If X_1, X_2, …, X_n are n random variables and a_1, a_2, …, a_n are constants, then:

$$E(a_1X_1 + a_2X_2 + \ldots + a_nX_n)$$
$$= a_1E(X_1) + a_2E(X_2) + \ldots + a_nE(X_n)$$

$$\mathrm{E}\left(\sum_{i=1}^{n} \mathrm{a}_{\mathrm{i}} X_i\right) = \sum_{i=1}^{n} \mathrm{a}_{\mathrm{i}} E\left(X_i\right)$$

Corollary: If X, and Y are random variables, then:

$E(X \pm Y) = E(X) \pm E(Y)$

Theorem 5:

If X is a random variable with variance $Var(X) = \sigma_X^2$

and if a and b are constants, then:

$$Var(aX \pm b) = a^2 \; Var(X)$$

$$\sigma_{aX \pm b}^2 = a^2 \, \sigma_X^2$$

Theorem 6:

If X_1, X_2, …, X_n are n <u>independent</u> random variables and a_1, a_2, …, a_n are constants, then:

$$Var(a_1X_1+a_2X_2+\ldots+a_nX_n) = Var(X_1)+ Var(X_2)+\ldots+ Var(X_n)$$

$$Var\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 Var(X_i)$$

$$\sigma^2_{a_1X_1+a_2X_2+\ldots+a_nX_n} = a_1^2\sigma^2_{X_1} + a_2^2\sigma^2_{X_2} + \ldots + a_n^2\sigma^2_{X_n}$$

Corollary:

If X, and Y are independent random variables, then:

$$Var(aX + bY) = a^2\ Var(X) + b^2\ Var\ (Y)$$

$$Var(aX - bY) = a^2\ Var(X) + b^2\ Var\ (Y)$$

$$Var(X \pm Y) = Var(X) + Var\ (Y)$$

Example 8:

Let X, and Y be two independent random variables such that $E(X) = 2$, $Var(X) = 4$, $E(Y) = 7$ and $Var(Y) = 1$.

Find:

1. $E(3X+7)$ and $Var(3X+7)$
2. $E(5X+2Y-2)$ and $Var(5X+2Y-2)$.
3. $E(5X-2Y-2)$ and $Var(5X-2Y-2)$.
4. $E(5x^2 -2y^2 -2)$.

Solution

1. $E(3X+7) = 3E(X)+7 = 3(2)+7 = 13$

& $Var(3X+7)= (3)^2\ Var(X)=(3)^2\ (4) =36$

2. $E(5X+2Y-2)= 5E(X) + 2E(Y) -2= (5)(2) + (2)(7) - 2= 22$

& $Var(5X+2Y-2)= Var(5X+2Y)= 5^2\ Var(X) + 2^2\ Var(Y)$

$= (25)(4)+(4)(1) = 104$

3. $E(5X-2Y-2)= 5E(X) - 2E(Y) -2= (5)(2) - (2)(7) - 2= -6$

& $Var(5X-2Y-2)= Var(5X-2Y)= 5^2\ Var(X) + 2^2\ Var(Y)$

$= (25)(4)+(4)(1) = 104$

Binomial Distribution

Bernoulli trial:

Bernoulli trial is an experiment with only two possible outcomes.

success (s) and failure (f)

The probability of success is $P(s) = p$

and the probability of failure is $P(f) = q = 1 - p.$

Binomial Random Variable

Consider the random variable:

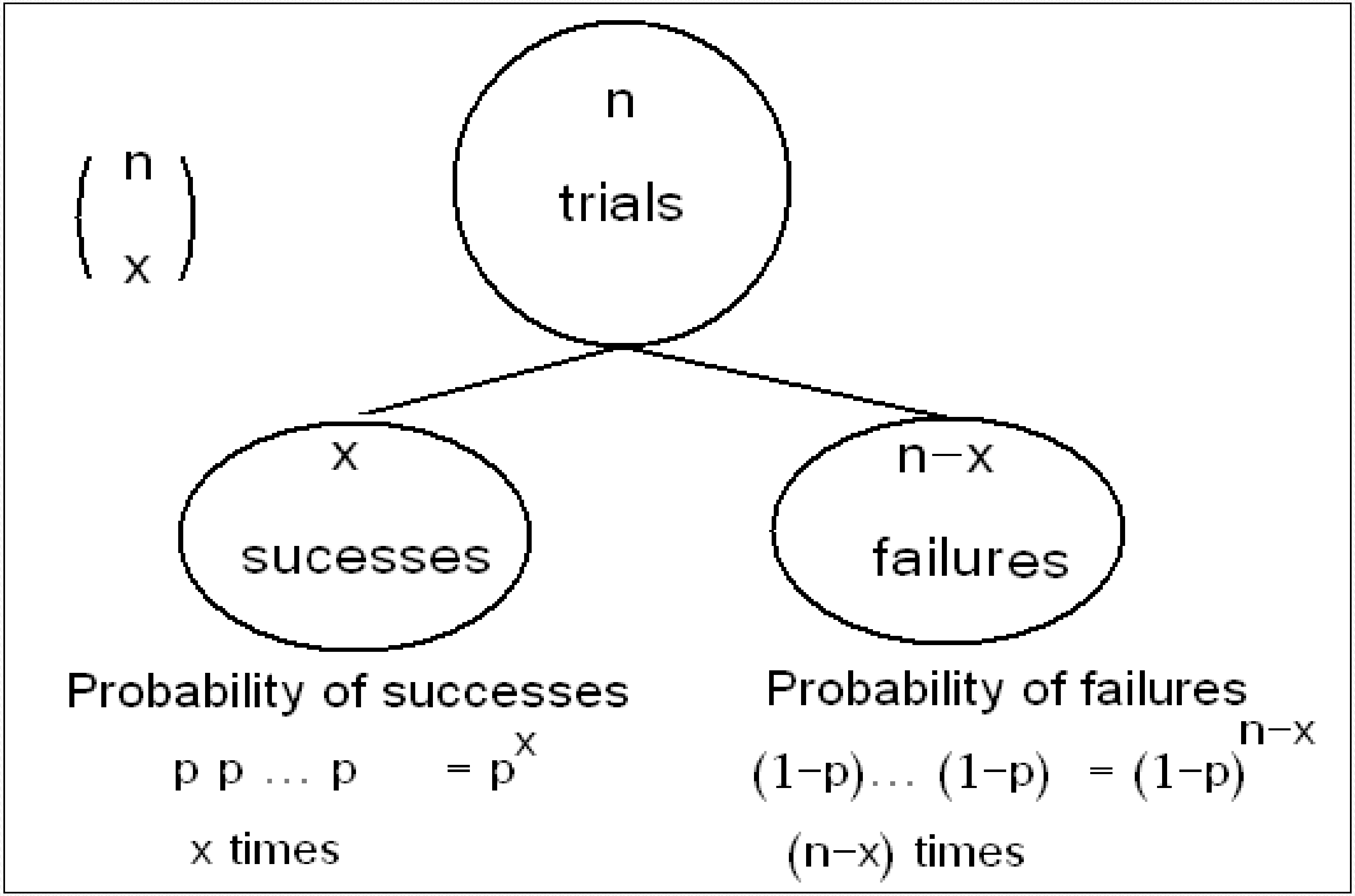
X = the number of successes in the n trials in a Bernoulli Process

The random variable X has a binomial distribution with parameters n (number of trials) and p (probability of success), and we write:

$$X \sim \text{Binomial}\,(n, p) \quad \text{or} \quad X \sim b\,(x; n, p)$$

The probability distribution of X is given by:

$$f(x) = P(X = x) = b(x; n, p) = \begin{cases} \dbinom{n}{x} p^x \, (1-p)^{n-x} \; ; & x = 0, 1, 2, \ldots, n \\ 0 \; ; & otherwise \end{cases}$$



$\binom{n}{x}$
n
trials
x
sucesses
n−x
failures
Probability of successes
$p\ p\ \ldots\ p \quad = p^x$
x times
Probability of failures
$(1-p)\ldots(1-p) \quad = (1-p)^{n-x}$
(n−x) times

We can write the probability distribution of X as a table as follows.

X	$f(x)=P(X=x)=b(x;n,p)$
0	$\binom{n}{0} p^0 (1-p)^n$
1	$\binom{n}{1} p^1 (1-p)^{n-1}$
$\vdots$	$\vdots$
$n-1$	$\binom{n}{n-1} p^{n-1} (1-p)^1$
n	$\binom{n}{n} p^n (1-p)^0$
Total	1

Example 1 :

Suppose that 25% of the products of a manufacturing process are defective. Three items are selected at random, inspected, and classified as defective (D) or non-defective (N). Find the probability distribution of the number of defective items.

Solution

Experiment: selecting 3 items at random, inspected, and classified as (D) or (N). The sample space is

S= {DDD, DDN, DND, DNN, NDD, NDN, NND, NNN}

Let X = the number of defective items in the sample

We need to find the probability distribution of X.

$$f(x) = P(X = x) = b\left(x; 3, \frac{1}{4}\right)$$

$$= \binom{3}{x} (0.25)^3 (1-0.25)^{3-x}$$

X	$f(x)=P(X=x)=b\left(x;3,\frac{1}{4}\right)$
0	$\binom{3}{0}\left(\frac{1}{4}\right)^{0}\left(\frac{3}{4}\right)^{3}=\frac{27}{64}$
1	$\binom{3}{1}\left(\frac{1}{4}\right)^{1}\left(\frac{3}{4}\right)^{2}=\frac{27}{64}$
2	$\binom{3}{2}\left(\frac{1}{4}\right)^{2}\left(\frac{3}{4}\right)^{1}=\frac{9}{64}$
3	$\binom{3}{3}\left(\frac{1}{4}\right)^{3}\left(\frac{3}{4}\right)^{0}=\frac{1}{64}$
Total	1

Theorem:

The mean and the variance of the binomial distribution $b(x; n, p)$ are:

$$\mu = n\,p$$

$$\sigma^2 = n\,p\,(1 - p)$$

Example 2

In the previous example (25% of the products of a manufacturing process are defective. Three items are selected at random, inspected, and classified as defective (D) or non-defective (N)). Find the expected value (mean) and the variance of the number of defective items.

Solution:

X = number of defective items

We need to find $E(X) = \mu$ and $Var(X) = \sigma^2$

We found that $X \sim$ Binomial (n, p) = Binomial $(3, 1/4)$

$n = 3$ and $p = 1/4$

The expected number of defective items is

$E(X)=\mu = n\,p = (3)\,(1/4) = 3/4 = 0.75$

The variance of the number of defective items is

$Var(X) = \sigma^2 = n\,p\,(1 - p) = (3)\,(1/4)\,(3/4) = 9/16 = 0.5625$

Example 3

In the previous example, find the following probabilities:

(1) The probability of getting at least two defective items.

(2) The probability of getting at most two defective items.

Solution:

X ~ Binomial (3, 1/4)

$$f(x) = P(X = x) = b(x; 3, \frac{1}{4}) = \begin{cases} \binom{3}{x} (\frac{1}{4})^{x} \, (\frac{3}{4})^{3-x} & \text{for} \quad x = 0, 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

X	$f(x) = P(X = x) = b\left(x; 3, \frac{1}{4}\right)$
0	$\binom{3}{0}\left(\frac{1}{4}\right)^{0}\left(\frac{3}{4}\right)^{3} = \frac{27}{64}$
1	$\binom{3}{1}\left(\frac{1}{4}\right)^{1}\left(\frac{3}{4}\right)^{2} = \frac{27}{64}$
2	$\binom{3}{3}\left(\frac{1}{4}\right)^{3}\left(\frac{3}{4}\right)^{1} = \frac{9}{64}$
3	$\binom{3}{3}\left(\frac{1}{4}\right)^{3}\left(\frac{3}{4}\right)^{0} = \frac{1}{64}$

1- The probability of getting at least two defective items:

$$P(X \geq 2) = P(X = 2) + P(X = 3) = f(2) + f(3) = \frac{9}{64} + \frac{1}{64} = \frac{10}{64}$$

2- The probability of getting at most two defective item:

$$P(X \leq 2) = P(X = 2) + P(X = 1) + P(X = 0)$$

$$= f(2) + f(1) + f(0) = \frac{9}{64} + \frac{27}{64} + \frac{27}{64} = \frac{63}{64}$$

or

$$P(X \leq 2) = 1 - P(X > 2) = 1 - P(X = 3) = 1 - \frac{1}{64} = \frac{63}{64}$$

Example 4

A package of 6 fuses are tested where the probability an individual fuse is defective is 0.05. (That is, 5% of all fuses manufactured are defective).

1- What is the probability one fuse will be defective?

2- What is the probability at least one fuse will be defective?

3- Find mean and standard deviation ?

n = 6 p = 0.05 q = 1-0.05 = 0.95

1- What is the probability one fuse will be defective?

$$f(x) = {}^{n}C_{x}\, p^{x} q^{(n-x)}$$

$$f(1) = {}^{6}C_{1}(0.05)^{1}(0.95)^{5} = 0.2321$$

2- What is the probability at least one fuse will be defective?

$$P(x \geq 1) = 1 - f(0)$$

$$1 - {}^{6}C_{0}(0.05)^{0}(0.95)^{6} = 0.2649$$

Poisson Distribution

If n independent trials, each of which results in a "success" with probability p, are performed, then when n is large and p small, the number of successes occurring is approximately a Poisson random variable with mean $\lambda = np$.

A random variable X, taking on one of the values 0, 1, 2, . . . , is said to be a Poisson random variable with parameter λ, $\lambda > 0$, if its probability mass function is given by

$$P\{X = x\} = \frac{e^{-\lambda}\lambda^{x}}{x!}, \qquad x = 0, 1, 2, \ldots$$

The symbol (e) stands for a constant approximately equal to 2.7183. It is a famous constant in mathematics, named after the Swiss mathematician L. Euler, and it is also the base of the so-called natural logarithm.

Some examples of random variables that usually obey, to a good approximation, the Poisson probability law are:

1. The number of misprints on a page (or a group of pages) of a book.
2. The number of people in a community living to 100 years of age.
3. The number of wrong telephone numbers that are dialed in a day.
4. The number of transistors that fail on their first day of use.
5. The number of customers entering a post office on a given day.

EXAMPLE 1

Suppose that the average number of accidents occurring weekly on a particular stretch of a highway equals 3. Calculate the probability that there is at least one accident this week.

SOLUTION

Let X denote the number of accidents occurring on the stretch of highway in question during this week. Because it is reasonable to suppose that there are a large number of cars passing along that stretch, each having a small probability of being involved in an accident, the number of such accidents should be approximately Poisson distributed. $\lambda = 3$

Hence, $P\{X \geq 1\} = 1 - P\{X = 0\}$

$$= 1 - \frac{e^{-3} 3^{0}}{0!} = 1 - e^{-3} \simeq 0.9502$$

EXAMPLE 2

The number of false fire alarms in a suburb of Houston averages 2.1 per day. Assuming that a Poisson distribution is appropriate, what is the probability that 4 false alarms will occur on a given day?

SOLUTION

The probability that 4 false alarms will occur on a given day is given by

$$P\{X = 4\} = \frac{e^{-(2.1)}(2.1)^4}{4!} \simeq 0.0992$$

EXAMPLE 3

Suppose the probability that an item produced by a certain machine will be defective is 0.01. Find the probability that a sample of 1000 items will contain at most one defective item. Assume that the quality of successive items is independent.

SOLUTION

Let X denote the number of defective items

$$\lambda = np = 1000 \times 0.01 = 10$$

$$P\{X \le 1\} = P\{X = 0\} + P\{X = 1\}$$

$$= \frac{e^{-10}10^0}{0!} + \frac{e^{-10}10^1}{1!} \simeq 0.0005$$

EXAMPLE 4

People enter a store, on average one every two minutes.

A. What is the probability that no people enter between 12:00 and 12:05?
B. Find the probability that at least 4 people enter during [12:00,12:05].

SOLUTION

A) Let X be the number of people that enter between 12:00 and 12:05. We model X as a Poisson random variable with parameter λ. Now, $\lambda = 2.5$, the average number of people that arrive in the 5-minute interval. But, if 1 person arrives every 2 minutes, on average (so 1/2 a person per minute), then in 5 minutes an average of 2.5 people will arrive. Thus,

$$P\{X = 0\} = \frac{e^{-(2.5)}(2.5)^0}{0!} = e^{-(2.5)}$$

B) $P\{X \geq 4\} = 1 - P\{X \leq 3\}$

$$= 1 - \left(P\{X = 0\} + P\{X = 1\} + P\{X = 2\} + P\{X = 3\}\right)$$

$$= 1 - \left(\frac{e^{-2.5}(2.5)^0}{0!} + \frac{e^{-2.5}(2.5)^1}{1!} + \frac{e^{-2.5}(2.5)^2}{2!} + \frac{e^{-2.5}(2.5)^3}{3!}\right)$$

EXAMPLE 5

A life insurance salesman sells on the average 3 life insurance policies per week. Use Poisson distribution to calculate the probability that in a given week he will sell

A. Some policies
B. 2 or more policies but less than 5 policies.
C. Assuming that there are 5 working days per week, what is the probability that in a given day he will sell one policy?

SOLUTION

(a) Let X be the number of policies sold in a week. Then

$$P\{X \geq 1\} = 1 - P\{X = 0\} = 1 - \frac{e^{-3}3^{0}}{0!} = 1 - e^{-3}$$

(b) We have

$$P\{2 \leq X < 5\} = P\{X = 2\} + P\{X = 3\} + P\{X = 4\}$$

$$= \frac{e^{-3}3^{2}}{2!} + \frac{e^{-3}3^{3}}{3!} + \frac{e^{-3}3^{4}}{4!} \simeq 0.61611$$

(c) Let X be the number of policies sold per day. Then

$\lambda = 3/5$. Thus,

$$P\{X = 1\} = \frac{e^{-3/5}(3/5)^{1}}{1!} \simeq 0.32929$$

EXAMPLE 6

At a checkout counter customers arrive at an average of 2 per minute. Find the probability that

A. At most 4 will arrive at any given minute.

B. At least 3 will arrive during an interval of 2 minutes.

C. 5 will arrive in an interval of 3 minutes.

D. Find mean and variance.

THANK *you*